



NYC Real Estate Analysis: Exploratory Data and Predictive Features

A focused examination of New York City housing data to identify key relationships and opportunities for predictive modelling

Focusing Our Lens on NYC

Rather than analysing the entire New York State dataset—which would introduce unnecessary complexity and regional variation—we've strategically narrowed our exploratory data analysis to New York City properties only.

This focused approach provides cleaner insights into the urban luxury market whilst reducing noise from diverse geographic areas with vastly different pricing dynamics.



Initial Findings: Bedrooms vs Price

Surprising Discovery

Our initial bar plot analysis revealed minimal correlation between bedroom count and property price—a counterintuitive finding that warranted deeper investigation.

Correlation Heatmap Insights

The correlation matrix uncovered meaningful relationships: price shows connections with bed count, bath count, and the brokerage handling the property.

Key Takeaway

These four variables—price, bedrooms, bathrooms, and brokered_by—emerge as promising features for predictive modelling and business intelligence applications.



When Bedroom Count Matters: Significant Price Differences

Post-hoc analysis using Tukey's HSD test reveals where bedroom count significantly impacts property valuation ($p < 0.001$):



1 vs 2 Bedrooms

Mean difference: ~£839,426



1 vs 3 Bedrooms

Mean difference: ~£2.5M



1 vs 4 Bedrooms

Mean difference: ~£5.1M



1 vs 16 Bedrooms

Mean difference: ~£24.1M

 **Key Insight:** Price escalation accelerates dramatically as properties transition from standard homes to luxury estates, with the most significant jumps occurring beyond 10 bedrooms.



Recommended Next Steps

01

Feature Engineering

Develop predictive models using our identified features: price, bed, bath, and brokered_by as primary variables.

02

Handle Statistical Noise

Address non-significant differences (e.g., fractional bedroom counts, categories with small sample sizes or high variance) through data cleaning and categorisation.

03

Segment Market Tiers

Create distinct models for standard residential (1-4 beds), upmarket properties (5-10 beds), and luxury estates (10+ beds) to capture tier-specific pricing dynamics.

04

Business Opportunities

Leverage insights to identify undervalued properties, optimise broker partnerships, and target high-margin luxury segments with precision.